

Physics 4140H: Quiz II

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Problem 1. Assume that the cost of flying an aircraft at height z is e^{-kz} per unit distance of flight-path, where k is a positive constant. Consider that the aircraft flies in the (x, z) -plane from the point $(-a, 0)$ to the point $(a, 0)$ where $z = 0$ corresponds to ground level, and where the z -axis points vertically upwards. Find the extrema for the problem. The cost integral to be minimized is

$$\int_{-a}^a e^{-kz} \sqrt{1 + (z')^2} dz.$$

Show that

$$z(x) = \frac{1}{k} \ln \left(\frac{\cos(kx)}{\cos(ka)} \right).$$

Use the boundary conditions given above to arrive at the answer.

Solution 1. Applying the Euler-Lagrange equation for $f = e^{-kz} \sqrt{1 + (z')^2}$,

$$\begin{aligned} \frac{\partial f}{\partial z} &= -ke^{-kz} \sqrt{1 + (z')^2} \\ \frac{\partial f}{\partial z'} &= e^{-kz} \frac{z'}{\sqrt{1 + (z')^2}} \\ \frac{d}{dx} \left(\frac{\partial f}{\partial z'} \right) &= -ke^{-kz} \frac{(z')^2}{\sqrt{1 + (z')^2}} + e^{-kz} \left[\frac{z'' \sqrt{1 + (z')^2} - \frac{(z')^2 z''}{\sqrt{1 + (z')^2}}}{1 + (z')^2} \right] \\ &= e^{-kz} \left[-\frac{k(z')^2}{\sqrt{1 + (z')^2}} + \frac{z''}{\sqrt{1 + (z')^2}} - \frac{(z')^2 z''}{[1 + (z')^2]^{3/2}} \right] \end{aligned}$$

which when plugged into the EL equation gives

$$-ke^{-kz} \sqrt{1 + (z')^2} = e^{-kz} \left[-\frac{k(z')^2}{\sqrt{1 + (z')^2}} + \frac{z''}{\sqrt{1 + (z')^2}} - \frac{(z')^2 z''}{[1 + (z')^2]^{3/2}} \right]$$

from which we can cancel the e^{-kz} terms as they are never zero, then multiply by $\sqrt{1 + (z')^2}$ to obtain

$$-k - \cancel{k(z')^2} = -\cancel{k(z')^2} + z'' - \frac{(z')^2 z''}{1 + (z')^2} \implies z'' - \frac{(z')^2 z''}{1 + (z')^2} + k = 0.$$

We then multiply through by the denominator of the remaining fraction to obtain

$$z''(1 + (z')^2) - (z')^2 z'' + k(1 + (z')^2) = 0 \implies z'' + \cancel{z''(z')^2} - \cancel{(z')^2 z''} + k + k(z')^2 = 0$$

which finally yields the expression

$$z'' + k(1 + (z')^2) = 0.$$

Now solving this took me a while, I went way off the right path at first and got stuck at a really horrible integral. The correct way to do this, I think (at least it gets me to the right answer) is to rewrite the primes in Leibnitz notation,

$$\frac{d}{dx} \frac{dz}{dx} + k \left(1 + \left(\frac{dz}{dx} \right)^2 \right) = 0$$

rearrange

$$\frac{d}{dx} \frac{dz}{dx} \frac{1}{1 + \left(\frac{dz}{dx}\right)^2} = -k$$

multiply by dx and integrate

$$\int \frac{1}{1 + \left(\frac{dz}{dx}\right)^2} d\frac{dz}{dx} = - \int k dx$$

and obtain

$$\int \frac{1}{1 + \left(\frac{dz}{dx}\right)^2} d\frac{dz}{dx} = -kx + C.$$

The integral of $1/(1+x^2)$ is doable by hand but we can get from a table that it comes out to $\arctan(x)$ so

$$\arctan\left(\frac{dz}{dx}\right) = -kx + C \implies \tan(C - kx)dx = dz \implies z(x) = -\frac{1}{k} \ln(\cos(C - kx))$$

where I've used the fact that the integral of $\tan(x)$ is $\ln(\sec(x)) = -\ln(\cos(x))$. Now we evaluate this between $(-a, 0)$ and $(x, 0)$ (where I've flipped the order because of the negative so we can get rid of it) to obtain, using $\ln(a) - \ln(b) = \ln(a/b)$,

$$z(x) = \frac{1}{k} \ln\left(\frac{\cos(C - kx)}{\cos(C + ka)}\right).$$

The boundaries require that $z(\pm a) = \frac{1}{k} \ln\left(\frac{\cos(C \pm ka)}{\cos(C + ka)}\right) = 0 \implies C = 0$ as $\ln(1) = 0$.